AI Research Collaboration Report

On 15 March 2025, Dr. Ananya Sharma joined the Artificial Intelligence Research Institute (AIRI) in Bengaluru as a senior researcher. AIRI was founded by Professor Rahul Mehta in 2018 and focuses on natural language processing, computer vision, and robotics.

Ananya Sharma previously worked at TechNova Solutions, a technology company headquartered in Mumbai. There, she collaborated with Dr. Vikram Patel on a project called DocSense, which automatically extracts entities and relationships from unstructured documents. Vikram Patel leads the NLP team at TechNova Solutions.

In June 2024, TechNova Solutions partnered with the Indian Institute of Science (IISc), located in Bengaluru, to develop better information extraction systems. Professor Rahul Mehta represented AIRI during the partnership negotiations. He believes that combining academic research with industrial experience will improve the accuracy of knowledge graph construction.

AIRI funded the initial phase of the project with ₹50 lakh. IISc provided the computing infrastructure, while TechNova Solutions contributed its document-processing software. The three organizations agreed to publish their findings in December 2025.

The DocSense project uses named entity recognition to identify people, organizations, and locations. It then applies dependency parsing and relation extraction techniques to discover connections between them. These extracted relationships are stored in Neo4j, where researchers can query the resulting knowledge graph.

Ananya Sharma will lead the next phase of the project. She will work with Vikram Patel and Rahul Mehta to evaluate the system on scientific papers and business reports. The team expects the improved system to reduce manual information extraction time by 40 percen